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Machine-learning surrogates for molecular hydrogen density in three-dimensional interstellar medium simulations using multi-scale spatial features

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ABSTRACT

Computing the molecular hydrogen number density n_{H_2} self-consistently in three-dimensional interstellar-medium (ISM) simulations requires solving stiff non-equilibrium chemistry coupled to radiative transfer, a cost that limits the resolution and parameter coverage of modern calculations. We evaluate machine-learning surrogates that predict $\log_{10}(n_{\mathrm{H}_2})$ cell-by-cell on a controlled suite of seven 128^3 UV-only chemistry simulations spanning $G_0 \in \{0.1, \ldots, 6.4\}$ Habing (14.7 million cells), under leave-one-G_0-out cross-validation: each fold withholds an entire simulation at a UV field strength absent from training. Alongside the usual log-space R^2 we evaluate the predictions as physical fields, decomposing the RMSE into bias and scatter and tracking the total H_2 mass budget and phase-conditional accuracy. These metrics expose what R^2 hides: models within 0.03 of each other in R^2 differ by factors of several in recovered H_2 mass. Multi-scale box-filter neighbourhood features (kernel widths 3, 5, 7 voxels; computed in $\approx$30 s by separable convolution) reduce the RMSE of gradient-boosted trees from 0.49 to 0.37 dex by acting on the scatter term, while a Ridge-stacked tree–MLP ensemble minimises squared error at the price of a systematic +0.3 dex offset that compounds to a $\times$1.7 mass over-prediction. Density-weighted training and a two-parameter, mass-weighted bias calibration fitted on training cubes alone repair the budget: the final pipeline reaches RMSE = 0.25 dex, $R^2 = 0.990$, and a total H_2 mass within 9 per cent of truth on every fold, with molecular-phase $R^2 = 0.99$. Ablating the solver-provided self-shielding factor costs little in aggregate (R^2 0.985 $\rightarrow$ 0.964) but most of the molecular-phase fidelity (molecular R^2 0.991 $\rightarrow$ 0.840). A 3D U-Net trained on the same suite is the best interpolator in the comparison (interior-fold $R^2 = 0.995$, RMSE 0.19 dex) but collapses at both extrapolation boundaries and over-predicts the H_2 mass of the hardest fold by more than two orders of magnitude, a failure mode the global calibration cannot repair. Within-cube sampling experiments show that reconstruction quality is controlled by sampling geometry rather than sample volume: 10 per cent uniformly random coverage yields $R^2 = 0.977$ while a contiguous half-cube fails entirely.

Key words: ISM: molecules – ISM: clouds – astrochemistry – methods: numerical – methods: statistical

1 INTRODUCTION

The formation of molecular hydrogen on dust-grain surfaces (Cazaux & Tielens 2004) and its photodissociation by Lyman–Werner UV photons (Habing 1968; Draine & Bertoldi 1996) together determine the molecular content of the interstellar medium (ISM). Because H_2 is both an important low-temperature coolant and the direct precursor of CO and the other molecular tracers used to infer star-formation rates (Narayanan et al. 2012), the H_2 abundance sets the initial conditions for the star-formation sequence. Computing it self-consistently within a three-dimensional hydrodynamics simulation requires integrating a stiff system of non-equilibrium chemistry equations at every grid cell, tracking the competition between H_2 formation on grain surfaces, UV photodissociation in the Lyman–Werner band (11.2–13.6 eV), and the H_2 self-shielding that occurs when H_2 columns attenuate their own dissociating radiation (Wolcott-Green et al. 2011). In simulations with 10^6–10^7 cells this chemistry solver can dominate the compute budget, and covering a wide UV-parameter range at full resolution is often infeasible (Grassi et al. 2014).

Two families of shortcuts exist. The older one replaces the solver with analytic or fitted formulae for the atomic-to-molecular transition, derived from photodissociation-region theory or calibrated on simulations (Krumholz et al. 2009; Gnedin & Kravtsov 2011; Sternberg et al. 2014; Bialy & Sternberg 2016); these are fast and interpretable, but each is tied to the equilibrium assumptions, geometry, and averaging scale for which it was derived. The newer family trains machine-learning (ML) surrogates on solver outputs: de Mijolla et al. (2019) emulate UCLCHEM abundances for molecular-line modelling, Holdship et al. (2021) advance a thermochemical state through a time-step with the Chemulator network, Grassi et al. (2022) compress a chemical network onto a learned low-dimensional manifold, and, closest to this work, Branca & Pallottini (2024) train DeepONet emulators on KROME (Grassi et al. 2014) integrations to reproduce the *time evolution* of a nine-species ISM photochemistry network at a single point, reaching mean relative errors below 3 per cent with a ∼128× speed-up over the stiff ODE solver. All of

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these emulators are zero-dimensional: they map a local initial state to a local final state and treat every cell independently. The present work addresses a complementary question that a zero-dimensional emulator cannot: how should a per-cell surrogate represent the *non-local* information — principally UV shielding by surrounding gas (cf. Glover & Mac Low 2007; Glover et al. 2010) — that couples a cell's equilibrium H_2 density to its three-dimensional environment, and how well do the resulting models generalise to radiation-field strengths absent from training?

Three properties of the problem shape the answer. First, n_{H_2} in our simulation suite spans roughly 16 orders of magnitude (from $\sim 10^{-12}$ to $\sim 10^4\,\mathrm{cm}^{-3}$), so all modelling is done in $\log_{10}$ space. Second, the atomic-to-molecular transition is sharp: the equilibrium abundance changes by orders of magnitude across a small range of shielding column. Third, and most importantly, a cell's chemical state is not a function of its local primitive variables alone; whether it is shielded from dissociating UV depends on the column of gas and dust between it and the radiation source.

A volumetric convolutional neural network (CNN) is the textbook answer to the non-locality problem: a 3D U-Net (Ronneberger et al. 2015; Çiçek et al. 2016) processing the whole volume can in principle learn shielding-like aggregations through its receptive field. But when the training set contains only a handful of simulation cubes at discrete parameter values, the volumetric sample count is tiny compared with the parameter count of even a small U-Net, and the behaviour of such models on out-of-distribution parameter values is not well characterised. We compare this learned approach against a deliberately simple engineered alternative: precomputed multi-scale box-filter averages of all input fields, appended to the per-cell feature vector of tabular models (gradient-boosted trees and multilayer perceptrons).

A second theme of the paper is that *how* a surrogate is scored determines what one learns about it. A predicted n_{H_2} field is a physical object: downstream applications integrate it into masses and column densities, split it into diffuse and molecular phases, and compare its statistics to observations. We therefore evaluate every model not only with the log-space R^2 conventional in this literature, but with a metric hierarchy designed around those uses — RMSE decomposed into bias and scatter, the total H_2 mass budget, phase-conditional errors, and distributional distances (Section 3.2). These metrics change the conclusions: the minimum-squared-error ensemble carries a systematic +0.3 dex offset, invisible at the precision R^2 is usually quoted, that compounds to a ×1.7 error in total H_2 mass. We treat that bias as a first-class modelling target, comparing two designed calibration functionals — one that centres the cell-mean log error and one that centres the mass budget — fitted only on training-cube quantities.

This paper makes five contributions. (i) We define a stringent out-of-distribution evaluation protocol — leave-one-G_0-out cross-validation over a controlled seven-simulation UV-parameter sweep, with every model scored on the same native 128^3 cells — together with a physics-weighted metric hierarchy, and show that both are needed: random cell-level splits would inflate every conclusion, and R^2 alone would miss the dominant error mode. (ii) We show that multi-scale spatial neighbourhood features deliver a consistent accuracy gain to tabular models at negligible cost, and that the gain is purely a scatter reduction — non-local context buys precision, not calibration. (iii) We identify the residual bias of the stacked ensemble as a smooth function of the UV field strength and repair it with a two-parameter calibration fitted on training cubes alone, comparing the cell-mean and mass-weighted variants; the final pipeline reaches RMSE = 0.25 dex, $R^2 = 0.990$, and a total H_2 mass within 9 per cent of truth on every fold. (iv) Because one input feature — the H_2 self-shielding factor f_{H_2} — is itself produced by the chemistry solver, we quantify the surrogate's dependence on it with a dedicated ablation, finding that the aggregate cost is modest (R^2 0.985 → 0.964 for the final model) but that the molecular-phase accuracy bears most of it (molecular R^2 0.991 → 0.840). (v) We characterise when the volumetric approach is and is not competitive: the 3D U-Net is the best interpolator in the comparison but degrades sharply at both extrapolation boundaries, over-predicts the H_2 mass of the hardest fold by more than two orders of magnitude, and is roughly two orders of magnitude more expensive to train.

The paper is structured as follows. Section 2 describes the data. Section 3 defines the evaluation protocol, metrics, and reproducibility measures. Section 4 describes the feature engineering, models, and calibration. Section 5 presents the main comparison, the mass-budget analysis, the f_{H_2} ablation, the volumetric baseline, and the error analysis of the final model. Section 6 reports supplementary experiments on cross-G_0 transfer and within-cube sampling geometry. Sections 7 and 8 give the discussion and conclusions.

2 DATA

2.1 Simulation suite

The data set consists of seven three-dimensional uniform-grid simulations from a UV-only chemistry suite. Each simulation occupies a $128 \times 128 \times 128$ grid of 2,097,152 cells, evolving identical turbulent ISM initial conditions under a spatially uniform, time-steady UV radiation field of strength G_0 in Habing units (Habing 1968). The seven runs differ in their imposed G_0 alone, sharing the same initial turbulent realisation and grid geometry; the suite is a controlled UV-parameter sweep, not a set of independent realisations. The evolved snapshots nonetheless differ in their hydrodynamic state as well as their chemistry — the median gas temperature rises from $\approx 1.0 \times 10^3$ K at $G_0 = 0.1$ to $\approx 1.1 \times 10^4$ K at $G_0 = 6.4$ as UV heating strengthens, with corresponding shifts in the density and velocity fields — while the large-scale structure remains strongly correlated across the suite. The G_0 values span a 64-fold range on a geometric sequence, $G_0 \in \{0.1, 0.2, 0.4, 0.8, 1.6, 3.2, 6.4\}$, and the snapshots represent quasi-equilibrium chemical states. The UV field illuminates each cube along the $+z$ axis, which breaks the full octahedral grid symmetry and dictates the augmentation strategy of Section 4.3.3. The dominant chemical processes in the suite are H_2 formation on grain surfaces (Cazaux & Tielens 2004) and Lyman–Werner photodissociation (Draine & Bertoldi 1996) attenuated by H_2 self-shielding computed along the illumination axis (Wolcott-Green et al. 2011). The total data set contains 14,680,064 cells.

We use this suite as a controlled testbed for surrogate methodology: the single-parameter design isolates the question of generalisation across the UV field strength from all other sources of variation. The corresponding limitation — that generalisation across turbulent realisations, metallicities, or density regimes is untested — is discussed in Section 7.5. A complete description of the simulation code, chemical network, and initial conditions (box size, resolution, driving, metallicity, dust model) is beyond the scope of this methods-focused paper and must accompany any future application of the surrogate to scientific inference; we state explicitly which conclusions depend only on the internal structure of the data (all of them) and which depend on the physical realism of the simulations (none directly).

Table 1. The 15 input features. "Transform" is the preprocessing applied before modelling. "Type" distinguishes local primitive variables, the global run parameter, and non-local quantities. "Deployable" indicates whether the feature would be directly available in a target application where the chemistry solver has not been run; the self-shielding factor f_{H_2} is the single problematic input and is treated by ablation in Section 5.3.

Feature	Units	Transform	Type	Deployable
n_H	cm^{-3}	$\log_{10}$	local	yes
T	K	$\log_{10}$	local	yes
n_{H^+}	cm^{-3}	$\log_{10}$	local	yes
`ext`	–	none	non-local proxy	yes
f_{H_2}	–	$\log_{10}$	non-local, solver	no
G_0	Habing	$\log_{10}$	global	yes
v_x, v_y, v_z	code	none	local	yes
B (6 face comps.)	code	none	local	yes

2.2 Physical fields

Each cell carries 18 raw fields: three integer grid coordinates (excluded from model input), the target n_{H_2}, and 15 input features summarised in Table 1. Fields spanning many orders of magnitude are log-transformed,

$$x_\ell = \log_{10}(x + \varepsilon), \qquad \varepsilon = 10^{-30}, \tag{1}$$

where ε guards against $\log(0)$ without affecting any physically realised value (the smallest non-zero n_{H_2} in the suite is $\sim 5\times10^{-12}$ cm^{-3} and the smallest f_{H_2} is $\sim 10^{-24}$).

The H_2 self-shielding factor f_{H_2} merits a precise statement. It is the attenuation factor applied to the Lyman–Werner photodissociation rate, computed by the chemistry solver from the H_2 column between each cell and the UV source. It is not algebraically derived from the cell's own n_{H_2} — we verified directly in the data that f_{H_2} does not equal the local molecular fraction $2n_{H_2}/(n_H + 2n_{H_2})$ for any cell, and that cells with nearly identical local n_{H_2} carry f_{H_2} values differing by orders of magnitude depending on their depth within the cloud — but as a column integral over the very field being predicted it imports target information through the solver, and it is an output of the very calculation the surrogate aims to replace. In a deployment scenario where the solver has not been run, f_{H_2} would have to be estimated, e.g. from a dust-extinction proxy or an approximate radiative-transfer sweep. Its contribution to surrogate accuracy is therefore quantified separately (Section 5.3) rather than assumed, and the configuration without it is the one we consider deployable. The dust-extinction proxy `ext` is also a column-type quantity, but unlike f_{H_2} it is computable from the density field alone without chemical information.

2.3 Target distribution

Figure 1 shows the distribution of $\log_{10}(n_{H_2})$ across G_0. The target spans $\approx$16 orders of magnitude, from $\sim 10^{-12}$ cm^{-3} in UV-exposed gas to $\sim 10^4$ cm^{-3} in shielded cores, making linear-space regression numerically useless and motivating log-space modelling throughout. At $G_0 = 0.1$ the distribution is bimodal — a UV-exposed population near $\log_{10}(n_{H_2}) \approx -10$ and a self-shielded molecular population near $\log_{10}(n_{H_2}) \approx 0$ — and the per-cube mean falls from -3.1 at $G_0 = 0.1$ to -7.2 at $G_0 = 6.4$ as photodissociation becomes dominant. This strong dependence of the marginal target distribution on G_0 is what makes leave-one-G_0-out evaluation demanding: each held-out cube has a target distribution shifted relative to everything seen in training.

3 EVALUATION PROTOCOL

3.1 Leave-one-G_0-out cross-validation

All headline results use leave-one-G_0-out seven-fold cross-validation: each fold withholds one entire cube ($\approx$2.1 M cells) and trains on the remaining six ($\approx$12.6 M cells). Random cell-level splits would place cells from every cube — hence every G_0 and the single shared turbulence realisation — in both training and test sets; under that protocol a pointwise gradient-boosted tree already exceeds $R^2 = 0.99$, telling us nothing about generalisation. The intended application is prediction at a UV field strength that has not been simulated, and leave-one-G_0-out measures exactly that. Every model — tabular and volumetric alike — is evaluated on the same native 128^3 cells of the held-out cube.

The two boundary folds require genuine extrapolation: no training data exist below $G_0 = 0.1$ or above $G_0 = 6.4$. The five interior folds require interpolation between bracketing cubes. Empirically, which folds are hardest depends on the model family (Table 2). Gradient-boosted trees lose most accuracy at the upward fold ($G_0 = 6.4$): tree leaves are piecewise-constant, so in the $\log G_0$ coordinate — the only input that identifies the regime — their splits saturate at the training range and the model cannot extrapolate the trend. The 3D U-Net degrades at both boundaries. The MLP and the ensembles built on it are instead weakest at the interior fold $G_0 = 0.8$, showing that G_0-extrapolation is not the only failure mode the protocol exposes.

3.2 Metrics

A predicted n_{H_2} field feeds downstream uses — mass budgets, phase statistics, synthetic observations — that are sensitive to different error properties, so we score every model with a fixed metric hierarchy rather than a single statistic. All metrics are computed in $\log_{10}(n_{H_2})$ space over the 128^3 cells of the held-out cube and aggregated over the seven folds as mean ± fold standard deviation; the seven folds, not the 14.7 million cells, are the inferential unit, and model-ranking statements are checked for per-fold consistency.

Primary metrics. The root-mean-square error (dex), decomposed exactly into a systematic and a random part,

$$\mathrm{RMSE}^2 = b^2 + s^2, \qquad b = \langle \hat{y} - y \rangle, \quad s = \mathrm{std}(\hat{y} - y), \tag{2}$$

where b is the mean residual (bias) and s the residual scatter; and the total H_2 mass ratio

$$\mathcal{M} = \frac{\sum_i 10^{\hat{y}_i}}{\sum_i 10^{y_i}}, \tag{3}$$

which on a uniform grid with equal cell volumes equals the ratio of predicted to true total H_2 mass. These two answer the question "can the predicted densities be trusted as physical values?": a bias of b dex that is invisible to variance-normalised statistics multiplies the mass budget by up to 10^b.

Secondary metrics. The log-space coefficient of determination

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \tag{4}$$

is retained for comparability with the emulator literature, with one caveat used throughout: because its denominator is the held-out fold's target variance, per-fold R^2 rewards models that win high-variance folds, and model rankings under mean R^2 can invert relative to pooled squared error (Section 5.1 contains a concrete example). We also report the fraction of cells with absolute residual below 0.1 and 0.3 dex ($f_{0.1}$, $f_{0.3}$); phase-conditional R^2 and bias for the molecular ($y > -4$) and diffuse ($y \le -4$) populations, the threshold lying in the

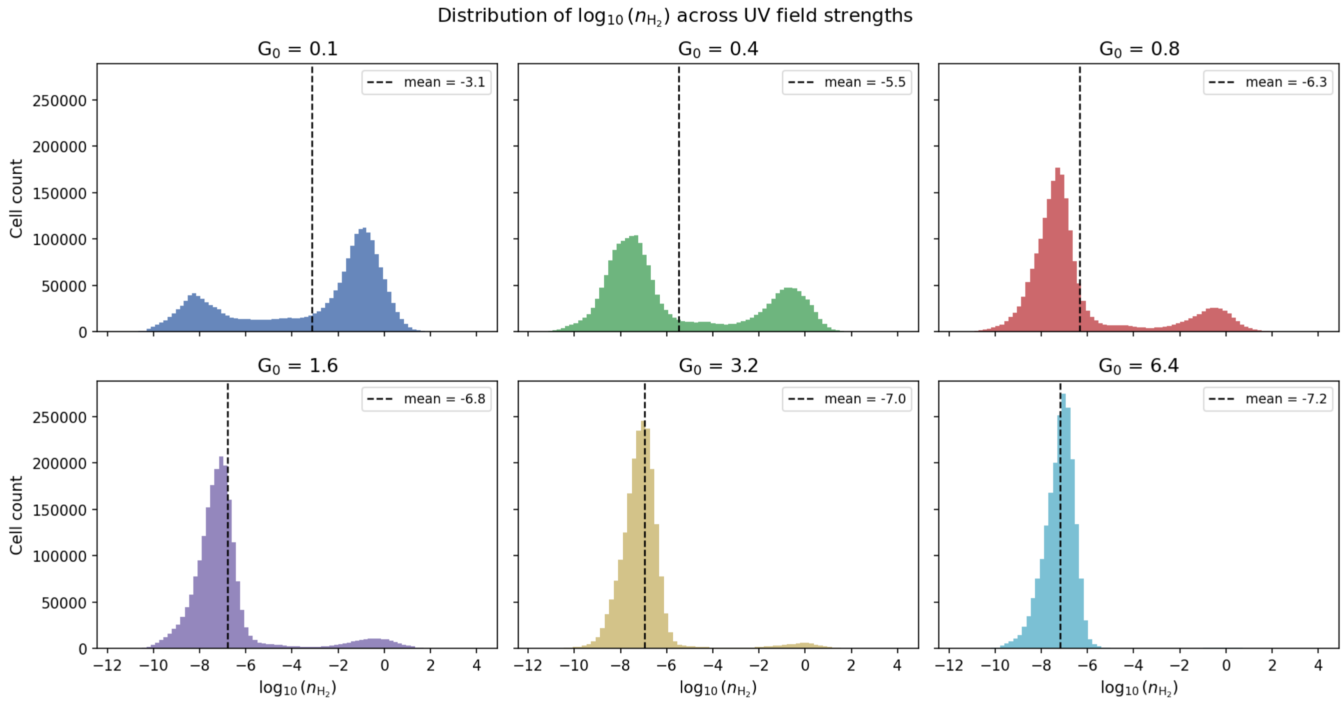

Figure 1. Distribution of $\log_{10}(n_{H_2})$ across six UV field strengths. The seventh cube, $G_0 = 0.2$, closely resembles $G_0 = 0.1$ and is omitted from this figure only to avoid visual redundancy; it is included in all quantitative analyses. The bimodal structure at low G_0 (UV-exposed and self-shielded populations) collapses to a single UV-dominated mode as G_0 increases. Dashed lines mark per-cube means. All modelling is performed in log-space.

gap between the two modes of Figure 1; and the mass-weighted mean absolute error $\mathrm{MAE}_{\mathrm{mw}}$ (weights $\propto 10^y$), which measures accuracy where the H_2 mass resides.

Supporting metrics. The 1-D Wasserstein distance W_1 (dex) between the true and predicted marginal distributions of $\log_{10}(n_{H_2})$ measures distributional realism independently of per-cell pairing; and a skill score

$$\mathrm{skill} = 1 - \mathrm{MSE}/\mathrm{MSE}_{\mathrm{ref}}, \tag{5}$$

with the pointwise 15-feature XGBoost model as the reference, replaces R^2's fold-mean denominator with a meaningful baseline.

Linear-space quantities require one numerical guard: before exponentiation, predictions are clipped to within 1 dex of the held-out fold's true range, preventing a handful of outlier cells from dominating $\mathcal{M}$ (and the linear-space R^2, which we report only in the field-level analysis of Section 5.5). For the well-calibrated tabular models the clip is essentially never active; for models that over-predict beyond the clip the reported $\mathcal{M}$ is a *lower bound* on the violation, and we flag those cases explicitly. The MLP's own predictions are additionally clamped to the training-target range ± 2 dex as part of the model definition (Section 4.3); this uses training information only.

3.3 Reproducibility

Every experiment is logged to a timestamped JSON file containing the full run configuration (feature list, hyperparameters, random seed, software environment) and per-fold metrics. Random seeds are fixed per fold (a global seed plus the fold index). The data files are fingerprinted by a SHA-256 checksum recorded in each log, so any change to the input data is detectable. Every result quoted in this paper is traceable to an archived log, and the complete pipeline runs end-to-end on one consumer GPU. The gradient-boosted-tree results are bit-identical across repeated runs of the pipeline; the neural models are seeded and reproducible on fixed hardware.

4 METHODS

4.1 Multi-scale spatial neighbourhood features

The dominant non-local effect on n_{H_2} is UV attenuation by surrounding gas. To give pointwise models access to spatial context without training a volumetric network, we precompute uniform box-filter averages of all 15 input fields. For each feature f and cube, the tabular values are reshaped to a 128^3 volume and filtered at kernel widths $k \in \{3, 5, 7\}$ voxels:

$$\bar{f}_{\boldsymbol{i},k} = \frac{1}{k^3} \sum_{\boldsymbol{j} \in \mathcal{N}_k(\boldsymbol{i})} f_{\boldsymbol{j}}, \tag{6}$$

where $\mathcal{N}_k(\boldsymbol{i})$ is the $k \times k \times k$ neighbourhood centred on $\boldsymbol{i}$ (kernel volumes of 27, 125, and 343 voxels); neighbourhoods are reflected at the box faces. The box filter is separable, so the cost is $O(N)$ per scale independent of k; computing all 45 filtered fields for all seven cubes takes ≈ 30 s. The filtered values are concatenated with the 15 local features into a 60-dimensional vector:

$$\boldsymbol{x}_i^{(60)} = \left[\boldsymbol{x}_i^{(15)} \,\|\, \bar{\boldsymbol{x}}_{i,3}^{(15)} \,\|\, \bar{\boldsymbol{x}}_{i,5}^{(15)} \,\|\, \bar{\boldsymbol{x}}_{i,7}^{(15)}\right]. \tag{7}$$

Figure 2 contrasts this representation with per-cell modelling. Spatial averages of the deployable column-type fields (density, extinction) act as crude shielding estimators; this is why the spatial features partially substitute for f_{H_2} in the ablation of Section 5.3.

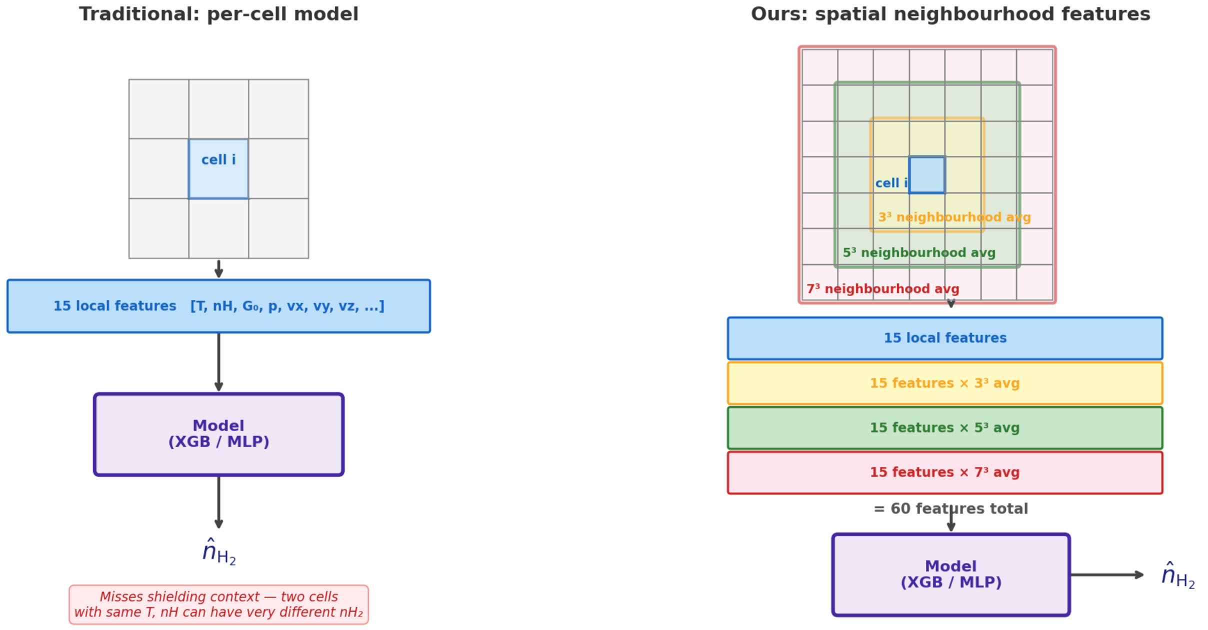

Figure 2. Schematic comparison of per-cell modelling (*left*) and the multi-scale spatial neighbourhood representation used here (*right*). Note that the 15-feature baseline already includes one non-local quantity, the solver-computed self-shielding factor f_{H_2}; what the spatial features add is explicit neighbourhood context for *all* fields, including the directly observable ones, at three nested scales.

4.2 Density-weighted training

The cell population is dominated by diffuse gas: at the phase threshold of Section 3.2 the molecular fraction of cells ranges from 0.8 per cent ($G_0 = 6.4$) to 68 per cent ($G_0 = 0.1$), while the H_2 *mass* resides almost entirely in the densest few per cent of cells. An unweighted log-space loss therefore optimises the diffuse majority and treats the mass-carrying tail as an afterthought. As a designed counterweight we train each tabular model in two variants: unweighted, and with smooth exponential sample weights

$$w(y) = \exp\big[\ln(\alpha)\, t(y)\big], \qquad t(y) = \mathrm{clip}\left(\frac{y - q_{99}}{q_{99.99} - q_{99}}, 0, 1\right), \quad (8)$$

with $\alpha = 100$ and q_p the p-th percentile of the training targets, normalised to unit mean so the effective learning rate is unchanged. The weight rises from 1× at the 99th percentile to 100× at the 99.99th: the mass-carrying tail is upweighted without destabilising the bulk fit. The percentile anchors and α are a designed choice, not tuned; their effect is reported in Section 5.1.

4.3 Models

4.3.1 Gradient-boosted trees

We use XGBoost (Chen & Guestrin 2016) with the GPU histogram method: 400 boosting rounds, maximum depth 6, learning rate 0.1, row subsampling 0.3, column subsampling 0.8. Depth variants 4 and 8 change the mean R^2 by less than 0.003 and were retired. Features are standardised per training fold (a no-op for trees in principle; retained for pipeline uniformity).

4.3.2 Wide MLP

The multilayer perceptron has hidden widths [512, 512, 256, 128] with batch normalisation (Ioffe & Szegedy 2015) and ReLU activations, trained with Adam (Kingma & Ba 2015) (learning rate 10^{-3}, weight decay 10^{-5}), batch size 262,144, 100 epochs per fold, cosine-annealed learning rate, and mixed precision. The full training fold is preloaded to GPU memory and batched by on-device permutation. Predictions are clipped to the training-target range ±2 dex: the surrogate is not permitted to invent densities far outside the training support (trees have this property inherently, their leaf values being bounded by training targets). Narrower and residual-block (He et al. 2016) variants give mean R^2 within ~0.02 with no consistent ordering across runs; differences between MLP architectures are comparable to seed-to-seed variation.

4.3.3 3D U-Net

The volumetric baseline is a 3D U-Net (Ronneberger et al. 2015; Çiçek et al. 2016) with three encoder stages, a bottleneck, trilinear-upsampling decoder, skip connections, and residual convolution blocks, trained and evaluated at the native 128^3 resolution (peak memory 5.7 GiB at 32 base channels and batch size 1 on an RTX 3090). Because each training sample is one entire volume (batch size 1), instance normalisation (Ulyanov et al. 2017) replaces batch normalisation; note that instance normalisation standardises each input volume by its own per-channel statistics, giving the network a per-cube input adaptation at test time that the tabular models do not receive (it uses input fields only, so no target information is involved). Channel inputs and the training target are standardised per fold from training-cube statistics; per-fold target standardisation is necessary for stable training because the target mean shifts by several dex from fold to fold (Section 2.3) and otherwise dominates the loss. Training uses Adam (learning rate 5×10^{-4}, weight decay 10^{-5}), gradient-norm clipping at 1.0, an unweighted mean-squared-error loss, and 150 epochs per fold.

Two protocol properties of the U-Net favour it relative to the tabular models and should be kept in mind when reading its results. First, the reported checkpoint is the epoch with the lowest loss on the held-out cube — a model-selection step the fixed-epoch tabular models do not get; the per-epoch training histories bound its advantage at ≤12 per cent in RMSE (final-epoch versus best-epoch validation loss ratios of 1.01–1.25 across folds and configurations), so U-Net numbers should be read as mild upper bounds. Second, the U-Net output is unbounded — unlike the range-clamped MLP and the inherently bounded trees — which matters for its mass-budget behaviour (Section 5.4). Convolutions use zero padding at the box faces.

Data augmentation respects the broken symmetry of the illuminated cube: of the 48 octahedral operations, only the eight z-preserving ones (rotations about z and reflections through planes containing z) are physically valid. Scalars transform by index permutation, velocities as polar vectors, and the face-centred magnetic field components as axial vectors with the appropriate face swaps. Using all 48 operations injects unphysical samples and degrades validation accuracy. Each fold therefore trains on $6 \times 8 = 48$ augmented volumes; the held-out cube is never augmented.

4.4 Ensembles and bias calibration

Two ensembles combine the spatial-feature XGBoost and MLP. The *equal-weight ensemble* averages their per-cell predictions. The *Ridge-stacked ensemble* fits a Ridge regression ($\alpha = 1$) meta-learner on out-of-fold (OOF) predictions of the training cubes and applies it to the held-out cube. In the model-comparison experiments the OOF predictions are reused from the outer cross-validation — the standard stacking shortcut, in which the base models that generate cube j's meta-training predictions were themselves trained with the held-out cube in their training set; with a two-coefficient meta-learner the indirect effect is small, but we do not describe those rows as fully nested. The deployed pipeline of Section 5.5 instead uses fully nested stacking: for each held-out cube, base models are retrained on five training cubes to generate the OOF predictions, and final base models on all six; the held-out cube is excluded throughout. The nested pipeline scores slightly *higher* than the shortcut rows, so the comparison-table stacking numbers are, if anything, conservative. The two base models have complementary inductive biases — piecewise-constant, conservative extrapolation for the trees; smooth interpolation for the MLP — and stacking them outperforms either alone in every experiment. Both the unweighted pair and the density-weighted pair (Section 4.2) are stacked, giving two stack families.

Bias calibration. The stacked predictions minimise squared error but are not guaranteed unbiased on a held-out G_0, and Section 5.2 shows they are not. As a designed audit step we therefore fit, for each held-out fold, the per-training-cube OOF residual of the stacked model against $\log_{10} G_0$ by two-parameter least squares, and subtract the fitted value at the held-out G_0 from the predictions. Only training-cube quantities enter the fit; the correction is a genuine out-of-sample prediction of the bias. The residual functional can be chosen in two ways, and the choice is a methodological decision we evaluate explicitly:

- *mean calibration* centres the unweighted per-cube mean log residual — the natural choice if all cells matter equally;
- *mass calibration* centres the mass-weighted mean residual (weights $\propto 10^y$), the constant-offset correction that centres the predicted *total H_2 mass*, since the same dense cells dominate both the weights and the mass budget.

If the model's residual were independent of density the two would coincide; their fitted offsets differ by 0.1–0.15 dex for the stacks (Section 5.2), and that gap is itself a useful one-number diagnostic of density-dependent error. Each stack family is reported raw, mean-calibrated, and mass-calibrated.

5 RESULTS

5.1 Main comparison

Tables 2 and 3 report the leave-one-G_0-out performance of all models on the full 15-feature input set (plus 45 spatial features where indicated), and Figure 3 shows the per-fold behaviour. The progression through the model families cleanly separates *what each ingredient buys*:

Spatial features buy precision. Adding the 45 neighbourhood features reduces XGBoost's RMSE from 0.49 to 0.37 dex and the MLP's from 0.52 to 0.32 dex, and the improvement is almost entirely in the scatter term of equation (2): XGBoost's residual scatter falls from 0.36 to 0.25 dex and the MLP's from 0.46 to 0.29 dex, while the bias terms move by $\lesssim 0.1$ dex. Non-local context sharpens the prediction; it does not recalibrate it.

Density weighting buys the mass budget — for free. The weighted tree variant is statistically indistinguishable from the unweighted one in R^2 (0.963 in both cases; per-fold values agree to ±0.0003) yet its fold mass ratios move from 0.47–0.93 to 0.89–1.03: upweighting the mass-carrying tail fixes *where* the model is accurate without degrading the bulk statistics. The weighted MLP behaves the same way (mass ratios 0.92–1.03, versus 0.94–1.16 unweighted, with $\mathrm{MAE}_{\mathrm{mw}}$ halved from 0.095 to 0.050 dex).

Stacking buys squared error and exposes a bias. The Ridge stacks have the lowest scatter in the comparison ($s = 0.25$–0.26 dex) but carry a systematic +0.32 dex mean offset on the held-out cubes — the meta-learner's intercept zeroes the residual on the pooled training folds, not on an unseen G_0. Section 5.2 treats this bias and its repair. After mass calibration the weighted stack is the best model in the comparison: RMSE = 0.294 dex (the lowest of all 17 evaluated variants), $R^2 = 0.985 \pm 0.015$, mass ratios 0.93–1.07 in every fold, molecular-phase $R^2 = 0.991$, and skill +0.55 against the pointwise-XGBoost reference. We adopt it as the final model.

Two methodological observations from Table 3 are worth recording because they generalise beyond this data set. First, the R^2 caveat of Section 3.2 is not hypothetical: the pointwise MLP *beats* pointwise XGBoost on mean R^2 (0.958 versus 0.950) while its pooled squared error is 18 per cent *worse* (skill −0.18) — the MLP wins the high-variance low-G_0 folds, which mean R^2 rewards. Second, W_1 and the threshold fractions expose a family difference that error moments hide: the trees are moderately wrong almost everywhere (compressed predicted distribution; $f_{0.3} = 0.50$, $W_1 = 0.30$), whereas the MLP is nearly right for most cells with a heavy error tail ($f_{0.3} = 0.73$, $W_1 = 0.22$, despite the larger RMSE). The best marginal realism in the table belongs to the weighted spatial MLP ($W_1 = 0.109$), slightly better than the stacks — Ridge averaging narrows the predicted distribution — which matters for applications that need statistically realistic fields rather than minimum per-cell error (Section 7).

5.2 The H_2 mass budget and bias calibration

The raw stacks illustrate why the mass budget is a primary metric. Their mean residual on held-out cubes is +0.32 dex — a factor of two in density — and the corresponding fold mass ratios are 1.08–1.65 (unweighted stack) and 1.65–1.84 (weighted stack): the minimum-squared-error combination over-predicts the total H_2 mass of an un-

Table 2. Leave-one-G_0-out log-space R^2 per held-out fold for all evaluated models, on the same native 128^3 cells. "+S" appends the 45 multi-scale spatial features (Section 4.1); "+W" uses density-weighted training (Section 4.2). Stack rows give the Ridge-stacked ensemble of the indicated base pair, raw and with the two bias calibrations of Section 4.4. The best value in each column is bold. Tabular rows are from the master comparison log; the U-Net row is its best configuration (32 base channels, 150 epochs, no dropout) trained at native 128^3.

Model	Mean R^2	σ	G_0=0.1	G_0=0.2	G_0=0.4	G_0=0.8	G_0=1.6	G_0=3.2	G_0=6.4
XGBoost	0.950	0.029	0.956	0.949	0.972	0.972	0.969	0.951	0.883
Wide MLP	0.958	0.028	0.927	0.947	0.924	0.940	0.986	0.992	0.992
XGBoost +S	0.963	0.033	0.976	0.982	0.985	0.982	0.975	0.956	0.885
Wide MLP +S	0.982	0.017	0.979	0.986	0.978	0.945	0.994	0.996	**0.997**
XGBoost +S+W	0.963	0.033	0.976	0.982	0.985	0.982	0.975	0.956	0.886
Wide MLP +S+W	0.981	0.020	0.976	0.986	0.981	0.935	0.994	0.996	0.996
Equal-weight ensemble (+S)	0.981	0.007	**0.983**	0.986	0.983	0.970	0.990	0.987	0.971
Stack (+S), raw	0.962	0.021	0.981	0.980	0.970	0.933	0.977	0.969	0.927
Stack (+S), mean-cal.	**0.985**	0.009	0.974	0.988	0.987	0.968	**0.995**	0.995	0.986
Stack (+S), mass-cal.	0.972	0.017	0.982	0.987	0.980	0.952	0.985	0.979	0.942
Stack (+S+W), raw	0.959	0.026	0.980	0.981	0.974	0.921	0.975	0.965	0.916
Stack (+S+W), mean-cal.	0.984	0.011	0.973	0.989	0.988	0.961	**0.995**	0.994	0.985
Stack (+S+W), mass-cal.	**0.985**	0.015	0.980	0.989	0.987	0.952	**0.995**	**0.996**	0.996
3D U-Net (32 ch.)	0.970	0.039	0.905	**0.993**	**0.995**	**0.996**	**0.995**	0.994	0.910

Table 3. Physical-field metrics for the models of Table 2 (cross-fold means; mass ratio $\mathcal{M}$ as the min–max range over folds). RMSE decomposes as $\mathrm{RMSE}^2 = b^2 + s^2$ (equation 2); $R^2_{\mathrm{mol}}/R^2_{\mathrm{dif}}$ are phase-conditional at $\log_{10}(n_{\mathrm{H_2}}) = -4$; $f_{0.1}$ is the fraction of cells within 0.1 dex; W_1 the Wasserstein distance between marginals; skill is measured against the pointwise XGBoost row (equation 5). The U-Net's mass range is a lower bound on its violation (predictions clipped at truth +1 dex; Section 3.2).

Model	RMSE (dex)	bias b (dex)	scatter s (dex)	$\mathcal{M}$ (range)	R^2_{mol}	R^2_{dif}	$f_{0.1}$	W_1 (dex)	skill
XGBoost	0.486	+0.24	0.36	0.51–0.90	0.978	0.68	0.20	0.30	0.00
Wide MLP	0.518	+0.21	0.46	1.00–2.59	0.981	0.57	0.56	0.22	−0.18
XGBoost +S	0.374	+0.20	0.25	0.47–0.93	0.988	0.81	0.21	0.25	+0.32
Wide MLP +S	0.322	+0.11	0.29	0.94–1.16	0.989	0.83	0.69	0.11	+0.46
XGBoost +S+W	0.374	+0.20	0.25	0.89–1.03	0.989	0.81	0.21	0.25	+0.32
Wide MLP +S+W	0.329	+0.10	0.30	0.92–1.03	0.989	0.82	0.70	0.11	+0.41
Equal-weight ensemble (+S)	0.314	+0.15	0.24	0.61–0.92	0.992	0.86	0.26	0.18	+0.55
Stack (+S), raw	0.424	+0.32	0.25	1.08–1.65	0.955	0.76	0.03	0.33	+0.10
Stack (+S), mean-cal.	0.295	−0.01	0.25	0.50–0.81	0.981	0.87	0.51	0.15	+0.60
Stack (+S), mass-cal.	0.359	+0.20	0.25	0.92–1.22	0.982	0.82	0.20	0.23	+0.35
Stack (+S+W), raw	0.433	+0.32	0.26	1.65–1.84	0.955	0.74	0.03	0.33	+0.02
Stack (+S+W), mean-cal.	0.301	−0.01	0.26	0.71–0.89	0.981	0.86	0.52	0.16	+0.57
Stack (+S+W), mass-cal.	0.294	+0.09	0.26	0.93–1.07	0.991	0.86	0.68	0.13	+0.55
3D U-Net (32 ch.)	0.310	−0.06	0.26	0.81–172	0.79	0.83	0.51	0.17	+0.48

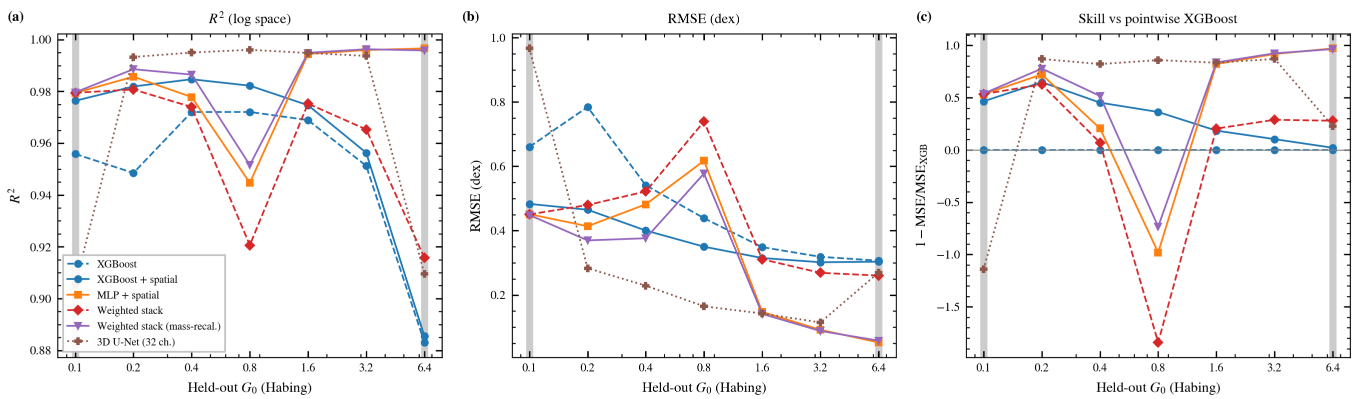

Figure 3. Per-fold behaviour of representative models across the G_0 sweep: log-space R^2 (*a*), RMSE (*b*), and skill against the pointwise XGBoost baseline (*c*). Grey bands mark the two extrapolation folds. The 3D U-Net (dotted) is the most accurate model on the interior folds and the least robust at both boundaries; the mass-calibrated weighted stack (purple) is the only model that remains strong across the entire range. XGBoost's decline with increasing G_0 reflects constant-valued tree extrapolation in the $\log G_0$ coordinate (Section 3.1).

seen cube by up to 80 per cent while scoring $R^2 > 0.95$ throughout. The offset varies smoothly with the held-out G_0, which is exactly the structure the two-parameter calibration of Section 4.4 is designed to remove; the fitted corrections are +0.2 to +0.4 dex depending on fold and functional.

The choice of residual functional matters, and Table 3 quantifies the designed comparison. *Mean* calibration centres the cell-mean bias essentially perfectly ($b = -0.01$ dex) and delivers the best skill (+0.60) — but *overshoots* the mass budget, dragging the fold mass ratios down to 0.50–0.81 for the unweighted stack: the cell-mean offset is set by the diffuse majority and is larger than the mass-weighted offset, so subtracting it pushes the dense cells too low. *Mass* calibration centres the budget ($\mathcal{M} = 0.93$–1.07 for the weighted stack) at the cost of a +0.09 dex residual cell-mean bias carried by the diffuse population. The two functionals differ because the stacks' residual error is density-dependent — a compression-to-the-mean pattern in which dense cells are under-predicted relative to diffuse ones — and the 0.1–0.15 dex gap between the fitted offsets measures that dependence directly. Under the primary-metric hierarchy (RMSE and mass budget) the mass-calibrated weighted stack is the best configuration; applications that weight all cells equally may prefer the mean-calibrated variant, and Table 3 provides both. Figure 4 shows the raw-versus-calibrated comparison for the deployed pipeline of Section 5.5, where the same picture holds fold by fold.

5.3 Dependence on the self-shielding factor

Because f_{H_2} is produced by the solver the surrogate aims to replace, Table 4 repeats the comparison with f_{H_2} excluded from both the local features and the spatial averages (14 local + 42 spatial features). In aggregate the surrogate degrades gracefully: the pointwise trees lose 0.039 in mean R^2 ($0.950 \rightarrow 0.911$), the pointwise MLP 0.049 (0.958 $\rightarrow$ 0.909), the spatial variants roughly half as much — as expected if neighbourhood averages of density and extinction act as crude column estimators — and the mass-calibrated weighted stack retains $R^2 = 0.964 \pm 0.014$ with its mass budget intact (fold mass ratios 0.93–1.16).

The phase-resolved metrics tell a sharper story that the R^2 comparison alone would miss: for the final model the ablation raises the RMSE by 60 per cent ($0.294 \rightarrow 0.469$ dex) and costs most of the molecular-phase fidelity — molecular R^2 falls from 0.991 to 0.840 and the molecular bias grows from -0.01 to -0.10 dex. The degradation concentrates precisely where H_2 matters most: at $G_0 = 0.1$, the most molecular cube, the final model's molecular R^2 drops to 0.70. The deployable (solver-independent) configuration is thus a usable surrogate for aggregate statistics and morphology, but applications requiring per-cell accuracy *inside* the molecular phase depend on a shielding estimate of f_{H_2} quality.

Feature attribution corroborates the ablation. Figure 5 shows XGBoost gain importances: log f_{H_2} dominates (0.58 of total gain for the pointwise model; 0.85 when its spatial averages are aggregated with it), followed by temperature, density, and G_0; the velocity and magnetic-field components contribute negligibly, suggesting a minimal six-feature deployable set that we have not separately evaluated.

For the volumetric model the available evidence is an upper bound of a different kind: a U-Net trained on only 11 inputs — temperature, G_0, velocity, and magnetic field, i.e. *no* chemistry-adjacent quantities at all (no n_H, n_{H^+}, extinction, or f_{H_2}) — still reaches $R^2 = 0.893 \pm 0.088$. Roughly 90 per cent of the log-density variance is thus recoverable from dynamics, thermodynamics, and the UV strength alone: the H_2 *morphology* is largely inferable without chemical inputs. The field-level metrics show what that number hides — residual scatter of 0.61 dex, only 17 per cent of cells within 0.1 dex, molecular-phase R^2 near zero on average and negative at both extrapolation folds, and fold mass ratios from 0.27 to a clipped ×800 — morphology yes, chemistry no.

5.4 The volumetric baseline

The 3D U-Net occupies a revealing position in Tables 2–3: it is simultaneously the best interpolator and the worst extrapolator in the comparison. On the five interior folds it is the most accurate model evaluated — $R^2 = 0.995$, RMSE 0.19 dex, beating the mass-calibrated weighted stack (0.984, 0.31 dex) on every one of them — direct evidence that a learned volumetric representation extracts non-local information the box-filter features do not. At the boundaries the picture inverts: mean edge-fold R^2 drops to 0.907 (RMSE 0.62 dex) against 0.988 (0.25 dex) for the calibrated stack, and at $G_0 = 0.1$ its skill against even the *pointwise* XGBoost baseline is -1.14: outside its training range the volumetric model is worse than the simplest tabular one.

The physical-field metrics make the extrapolation failure qualitative, not merely quantitative. At $G_0 = 6.4$ the U-Net's molecular-phase bias is +1.09 dex and its fold mass ratio exceeds 170 (a lower bound; its unbounded output is clipped at truth +1 dex before the mass integral, Section 3.2); at $G_0 = 0.1$ it under-predicts by -0.63 dex overall. Its interior-fold mass ratios are also uncontrolled (0.94–16.7), consistent with its unweighted, unbounded training. Crucially, this failure is *not* repairable by the calibration of Section 4.4: the U-Net's held-out bias is phase-localised (molecular +1.09 dex against a small diffuse bias at $G_0 = 6.4$) and wildly non-monotonic in G_0 (fold mass ratios 0.81, 3.9, 0.94, 1.8, 2.8, 16.7, 172), whereas the stack's bias is a global, G_0-smooth offset — the difference between a correctable systematic and a genuine extrapolation breakdown.

Two practical considerations reinforce the tabular default for suites of this size. *Configuration sensitivity*: in a dedicated configuration study conducted at reduced (64^3) resolution, U-Net mean R^2 ranged from 0.80 to 0.97 across capacity variants (16, 32, 64 base channels; 1.5 M–23 M parameters) and training protocols, with a dropout-regularised residual variant losing most of its accuracy at the extrapolation folds (0.87/0.82 versus 0.90/0.95 for the dropout-free baseline); with 48 augmented training volumes against millions of parameters, training noise is large, and any single U-Net number carries a wide implicit error bar. XGBoost results are bit-identical across runs and MLP variation stays within ~0.01. *Cost*: the seven-fold U-Net training takes ≈4.6 h on an RTX 3090, versus ≈1 h for the *entire* 17-variant tabular comparison including spatial-feature construction. We did not pursue CNN-specific remedies (density-weighted volumetric losses, deeper augmentation, pretraining) and therefore make no claim about the ultimate capability of volumetric architectures; the operational conclusion is that on a suite of this size the engineered spatial features achieve comparable interior accuracy with far greater out-of-distribution robustness at a small fraction of the cost.

5.5 Error analysis of the final model

For per-cell and field-level analysis we use the deployed pipeline: the density-weighted spatial stack with fully nested stacking and mass calibration (Section 4.4), which saves full 128^3 prediction volumes for each held-out fold. Its per-fold R^2 is 0.977, 0.988, 0.990, 0.987, 0.995, 0.996, 0.998 for $G_0 = 0.1$–6.4 (mean 0.990 ± 0.006; RMSE 0.25 dex; the nested protocol scores slightly above the comparison-table row, as noted in Section 4.4). The mass budget is closed in

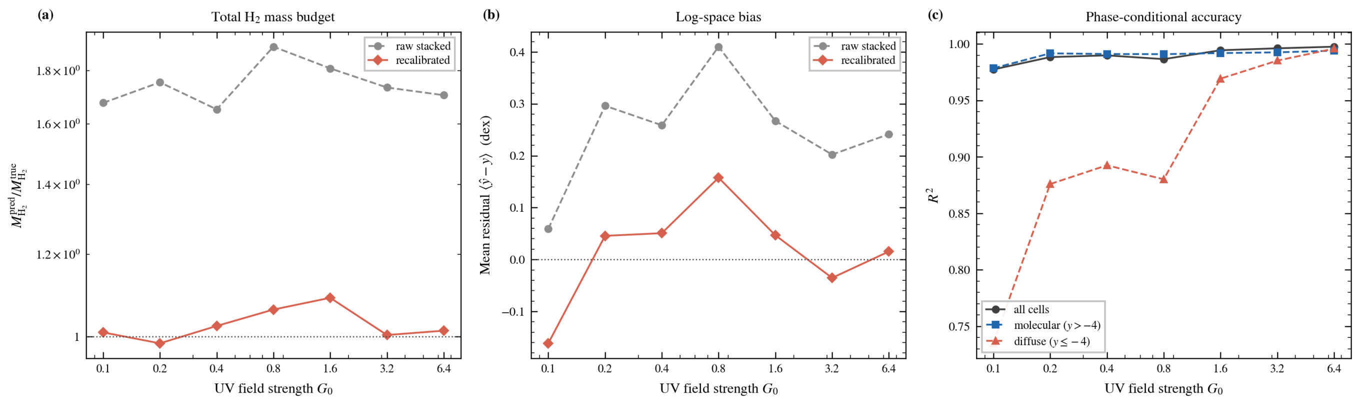

Figure 4. Mass budget and phase-conditional accuracy of the deployed (fully nested) pipeline of Section 5.5. *(a)* Total H_2 mass ratio per held-out fold: the raw stack (grey) over-predicts by $\times 1.65$–1.90 at every G_0; the mass-calibrated prediction (red) is within 9 per cent of the true mass in every fold. *(b)* Cell-mean log residual: the calibration also removes most of the mean bias. *(c)* Phase-conditional R^2: molecular-phase accuracy is uniformly high ($R^2 \geq 0.98$); the diffuse phase is hardest at low G_0, where UV-exposed gas spans the steep transition.

Table 4. Ablation of the solver-derived self-shielding factor f_{H_2}: key metrics with the full feature set (left of each arrow) and with f_{H_2} excluded from the local features and the spatial averages (right). R^2 entries give mean ± fold std.; RMSE and molecular-phase entries are cross-fold means. All runs use the identical pipeline, seeds, and hyperparameters; the without-f_{H_2} mass-calibrated stack keeps fold mass ratios of 0.93–1.16.

Model	R^2	RMSE (dex)	$R^2_{\rm mol}$
XGBoost	$0.950 \pm 0.029 \rightarrow 0.911 \pm 0.033$	$0.49 \rightarrow 0.70$	$0.978 \rightarrow 0.701$
Wide MLP	$0.958 \pm 0.028 \rightarrow 0.909 \pm 0.057$	$0.52 \rightarrow 0.76$	$0.981 \rightarrow 0.713$
XGBoost +S	$0.963 \pm 0.033 \rightarrow 0.941 \pm 0.029$	$0.37 \rightarrow 0.54$	$0.988 \rightarrow 0.815$
Wide MLP +S	$0.982 \pm 0.017 \rightarrow 0.953 \pm 0.027$	$0.32 \rightarrow 0.55$	$0.989 \rightarrow 0.809$
Stack (+S+W), mass-cal.	$0.985 \pm 0.015 \rightarrow 0.964 \pm 0.014$	$0.29 \rightarrow 0.47$	$0.991 \rightarrow 0.840$

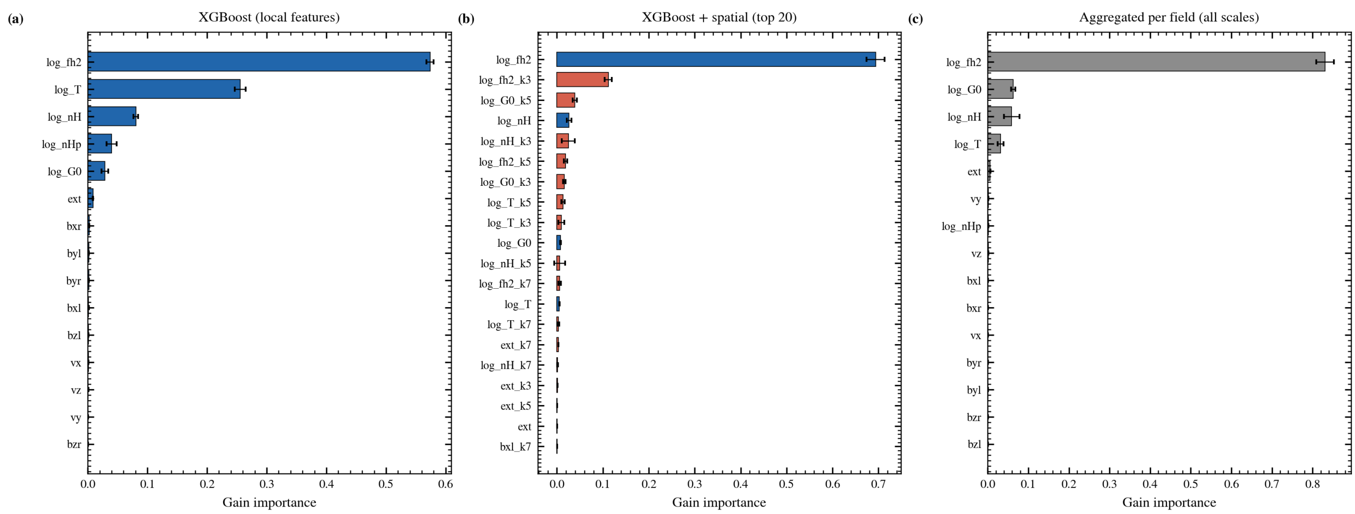

Figure 5. XGBoost gain importance, mean ± std over the seven folds. *(a)* Pointwise model (15 local features): the solver-derived log f_{H_2} dominates, followed by temperature and density. *(b)* Spatial-feature model, top 20 of 60 features (blue = local, red = neighbourhood means): f_{H_2} and its spatial averages fill the top ranks. *(c)* Importance aggregated per base field over all scales. Velocity and magnetic-field components are negligible in every view.

every fold ($\mathcal{M} = 0.985$–1.090), the molecular-phase R^2 is 0.979–0.994, the cell-mean bias is +0.02 dex, and the mass-weighted MAE is 0.06 dex. Figures 4 and 6–10 are derived from these volumes.

Figure 6 shows predicted versus true $\log_{10}(n_{H_2})$ for all seven folds: the model reproduces the bimodal structure in every fold, with scatter concentrated in the diffuse UV-exposed population and a tight locus along the identity for the shielded population. Figure 7 shows the residual distributions: narrow cores (widths 0.04–0.44 dex, shrinking monotonically with G_0) with per-fold means between −0.16 and +0.16 dex. The residual structure identifies $G_0 = 0.8$ as the hardest fold for this pipeline — its bias (+0.16 dex) and its fraction of cells within 0.1 dex (38 per cent, against 60–99 per cent elsewhere) are the weakest in the suite even though its R^2 is 0.987 — a reminder that interior folds are not automatically easy.

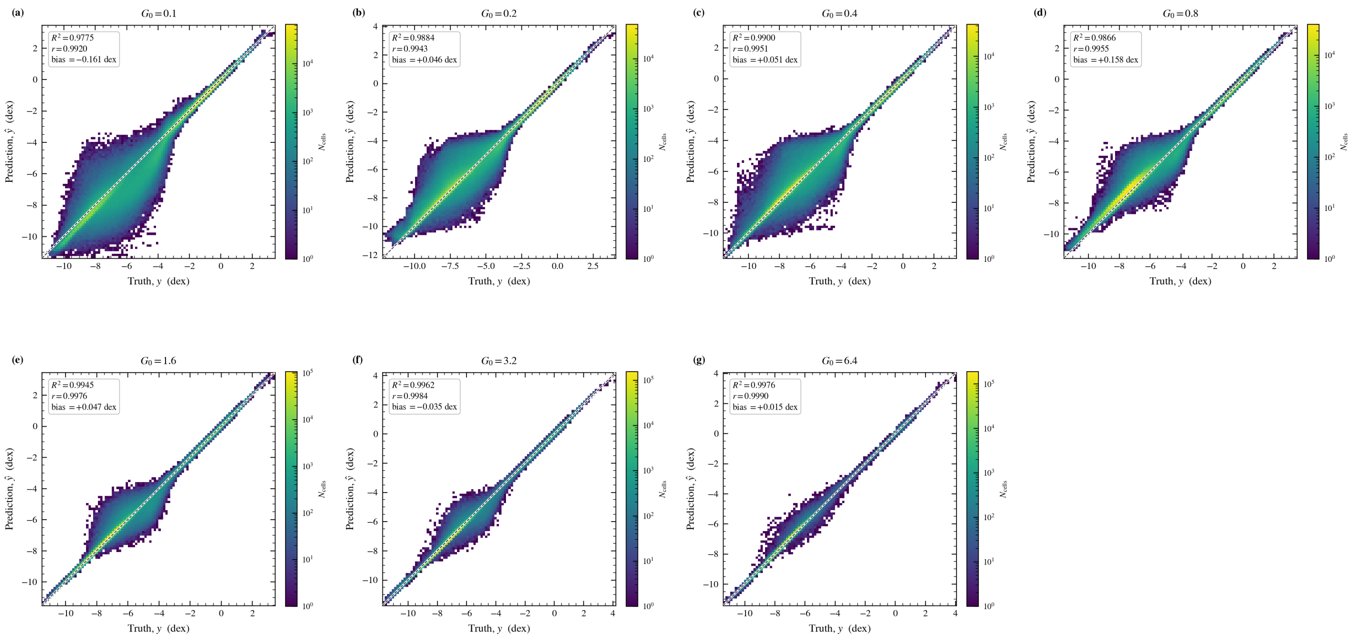

Figure 6. Predicted versus true $\log_{10}(n_{\rm H_2})$ per held-out fold (two-dimensional histograms, log colour scale; dashed line = identity), for the deployed mass-calibrated pipeline. Per-fold R^2, Pearson r, and bias are annotated. The bimodal target structure is reproduced in every fold; scatter is dominated by the diffuse UV-exposed population.

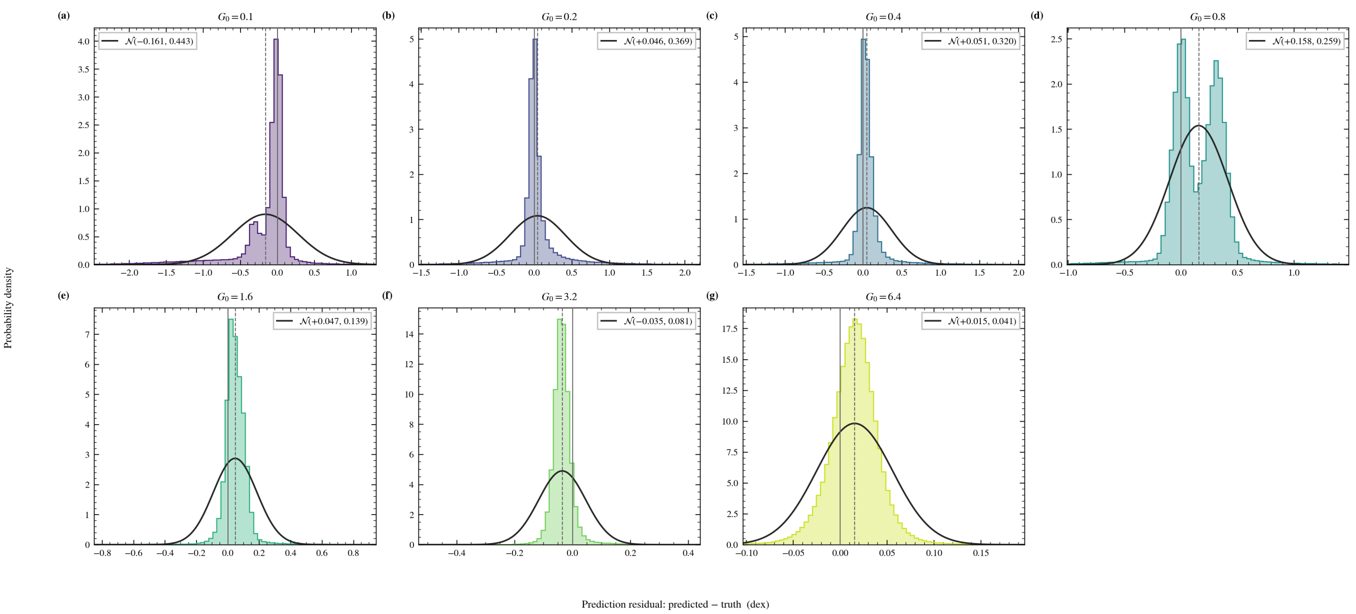

Figure 7. Residual distributions (predicted − true, dex) per fold, with best-fit normal curves and their (μ, σ) annotated. Distribution widths decrease monotonically with G_0; per-fold means lie between -0.16 and $+0.16$ dex after mass calibration. The multi-modal structure at several folds means the Gaussian curves summarise location and scale only and should not be read as a claim of normality.

Figure 8 stratifies the errors by gas density. The largest mean absolute errors occur in the *low*-density deciles of the low-G_0 folds, where small shielding-column differences move cells across the steep atomic-to-molecular transition; this is the same diffuse-phase weakness seen in the phase-conditional panel of Figure 4 (diffuse R^2 of 0.74 at $G_0 = 0.1$, rising to ≥ 0.97 for $G_0 \geq 1.6$). Conversely, restricting the evaluation to the densest cells *lowers* R^2 (panels a and d) even though absolute errors there are small: within the shielded cores the target variance is small, so R^2 — a variance-normalised metric — is a harsh statistic in the high-density tail. Both views are shown to avoid the misleading impression either one gives alone. Figure 9 compares predicted and true marginal distributions per fold: the bimodal shape and peak locations are reproduced, with W_1 distances of 0.02–0.17 dex (mean 0.08).

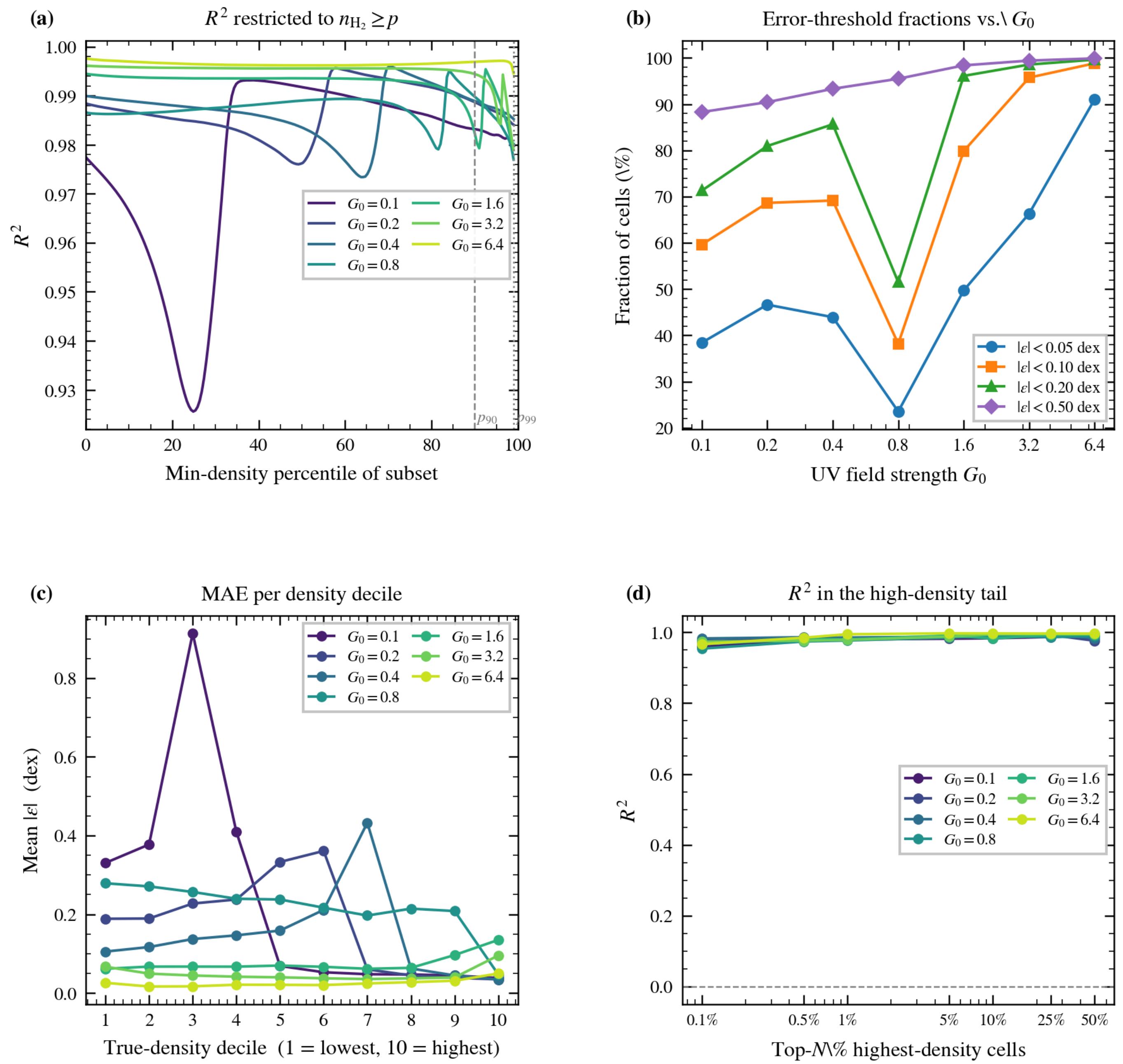

Figure 8. Density-stratified error analysis of the deployed pipeline. *(a)* R^2 on the subset of cells above a minimum-density percentile: R^2 *decreases* toward the dense tail because the within-subset target variance shrinks faster than the errors. *(b)* Fraction of cells with absolute residual below fixed thresholds versus G_0; the dip at $G_0 = 0.8$ reflects that fold's residual +0.16 dex bias. *(c)* MAE per true-density decile: errors peak in the low-density deciles of the low-G_0 folds, at the steep atomic-to-molecular transition; high-G_0 folds are flat at $\lesssim 0.1$ dex. *(d)* R^2 within the top-N per cent densest cells.

Figure 10 shows mid-plane slices of truth, prediction, and absolute error. At low G_0 the field is rich in filamentary shielded structure and the residuals concentrate at cloud boundaries — the spatial expression of the transition-region errors of Figure 8 — while at high G_0 the field is nearly uniform and residuals are small and unstructured.

6 SUPPLEMENTARY EXPERIMENTS

6.1 Single-cube transfer across G_0

To measure how far the learned mapping transfers across the UV parameter, we train the weighted spatial stack on each individual cube and predict all seven, producing a 7×7 matrix of log-space R^2 (Figure 11). In-sample (diagonal) fits average $R^2 = 0.997$. Transfer degrades systematically with G_0 separation: averaged over the matrix, one geometric step ($\times 2$ in G_0) gives $R^2 = 0.966$, two steps 0.888, three steps 0.729, four steps 0.452, five steps 0.158, and the

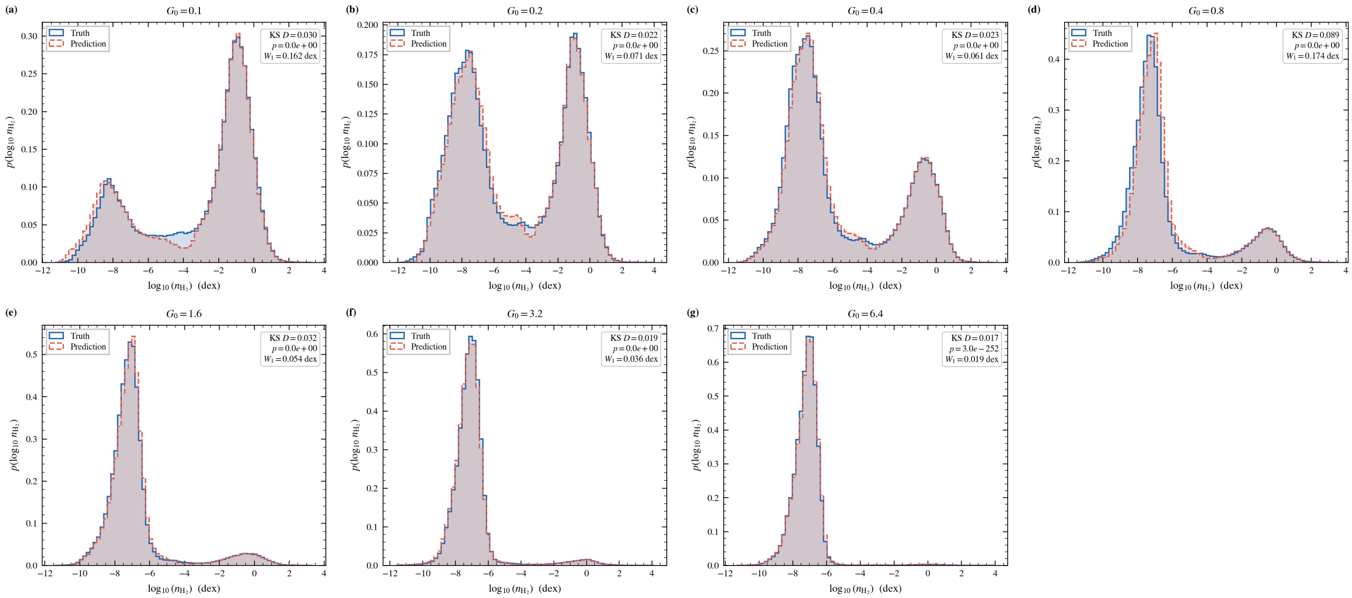

Figure 9. True (filled) versus predicted (outline) marginal distributions of $\log_{10}(n_{\mathrm{H_2}})$ per fold for the deployed pipeline, annotated with the two-sample Kolmogorov–Smirnov statistic and the Wasserstein distance W_1 (dex). The bimodal structure and peak positions are reproduced in every fold.

full 64-fold separation 0.006, with substantial spread between directions and cube pairs at the larger separations. The mean off-diagonal R^2 is 0.706. The learned local mapping is thus approximately universal within a factor of ∼2–4 in G_0 and degrades beyond it, which (i) validates leave-one-G_0-out as a non-trivial protocol — each held-out cube is one step from its nearest training neighbour, so single-cube knowledge alone would cap performance near 0.97, below what the six-cube training sets achieve — and (ii) confirms that none of the seven G_0 values is redundant in the training set.

6.2 Within-cube sampling geometry

We next ask how reconstruction of a *single* cube's $n_{\mathrm{H_2}}$ field depends on which cells are used for training — a question about sampling geometry in the simulation volume, not (we emphasise) a claim about observational survey design, since real observations deliver projected, beam-convolved, noisy quantities rather than simulation fields, and $f_{\mathrm{H_2}}$ is not an observable. For each cube we train the weighted spatial stack on a subset of cells and evaluate on the complement, comparing uniformly random subsets (1–75 per cent), contiguous half-cube slabs along each axis, and compact cubic subregions (1–50 per cent).

In these experiments the spatial features must respect the partial coverage. Field volumes are masked before filtering: cells outside the training subset are set to zero, and the box filter of equation (6) is applied to the masked volume,

$$\bar{f}^{\mathrm{obs}}_{\boldsymbol{i},k} = \frac{1}{k^3} \sum_{\boldsymbol{j} \in \mathcal{N}_k(\boldsymbol{i})} m_{\boldsymbol{j}} f_{\boldsymbol{j}}, \qquad m_{\boldsymbol{j}} = \begin{cases} 1 & \boldsymbol{j} \in \text{training subset}, \\ 0 & \text{otherwise}, \end{cases} \tag{9}$$

with *no* renormalisation by the local sample count $\sum_j m_j$: a neighbourhood with no sampled cells yields zero, and a sparsely sampled neighbourhood yields its sampled-cell sum diluted by the full kernel volume. The local (unfiltered) features of test cells use their true values; only the target labels and the neighbourhood context are restricted to the training subset. This convention treats the surrounding field values as unknown wherever they were not sampled; it is deliberately conservative — training cells near a subset boundary also receive degraded context — and its dilution property matters for interpreting the sparse-coverage results below.

Figure 12 and the run statistics give a consistent picture across all seven cubes. Uniformly random coverage of 10 per cent or more reconstructs the field accurately (test R^2 mean over cubes: 0.977 at 10 per cent, 0.982 at 25, 0.986 at 50); 5 per cent gives 0.902. At 1 per cent random coverage the expected number of sampled cells inside even a 7^3 kernel is only ∼3.4, the diluted spatial features carry little signal, and the outcome is unstable: mean R^2 of +0.24 with a range of −1.74 to +0.90 across cubes. Contiguous geometries show the opposite failure mode: a half-cube slab — 50 per cent of the volume — fails on every axis (mean R^2 of −0.65, −0.25, −0.25 for x, y, z slabs, with minima as low as −5.1), because the unsampled half contains physical regimes entirely absent from training. Compact boxes are intermediate and nearly flat in coverage (mean $R^2 \approx 0.76$–0.86 from 1 to 50 per cent). Notably, at 1 per cent coverage a compact box *outperforms* random sampling (0.76 versus 0.24): inside the box the masked spatial features are exact, whereas under sparse random sampling they are uniformly diluted. Geometry, not volume, is the controlling variable, and the best geometry depends on the coverage budget: contiguous regions win when coverage is very sparse, uniform sampling wins everywhere above a few per cent.

7 DISCUSSION

7.1 Feature engineering versus architecture

On this suite, input representation dominates architecture choice among the tabular models. The spatial features cut the RMSE of both tabular families by 23–38 per cent — an improvement several times larger than any architecture variation we tested (tree depth, MLP width and depth, residual connections, all $\lesssim 0.01$ in R^2 with no consistent ordering) — and stacking plus calibration buys a further ∼9 per cent of RMSE over the best single model. This is consistent

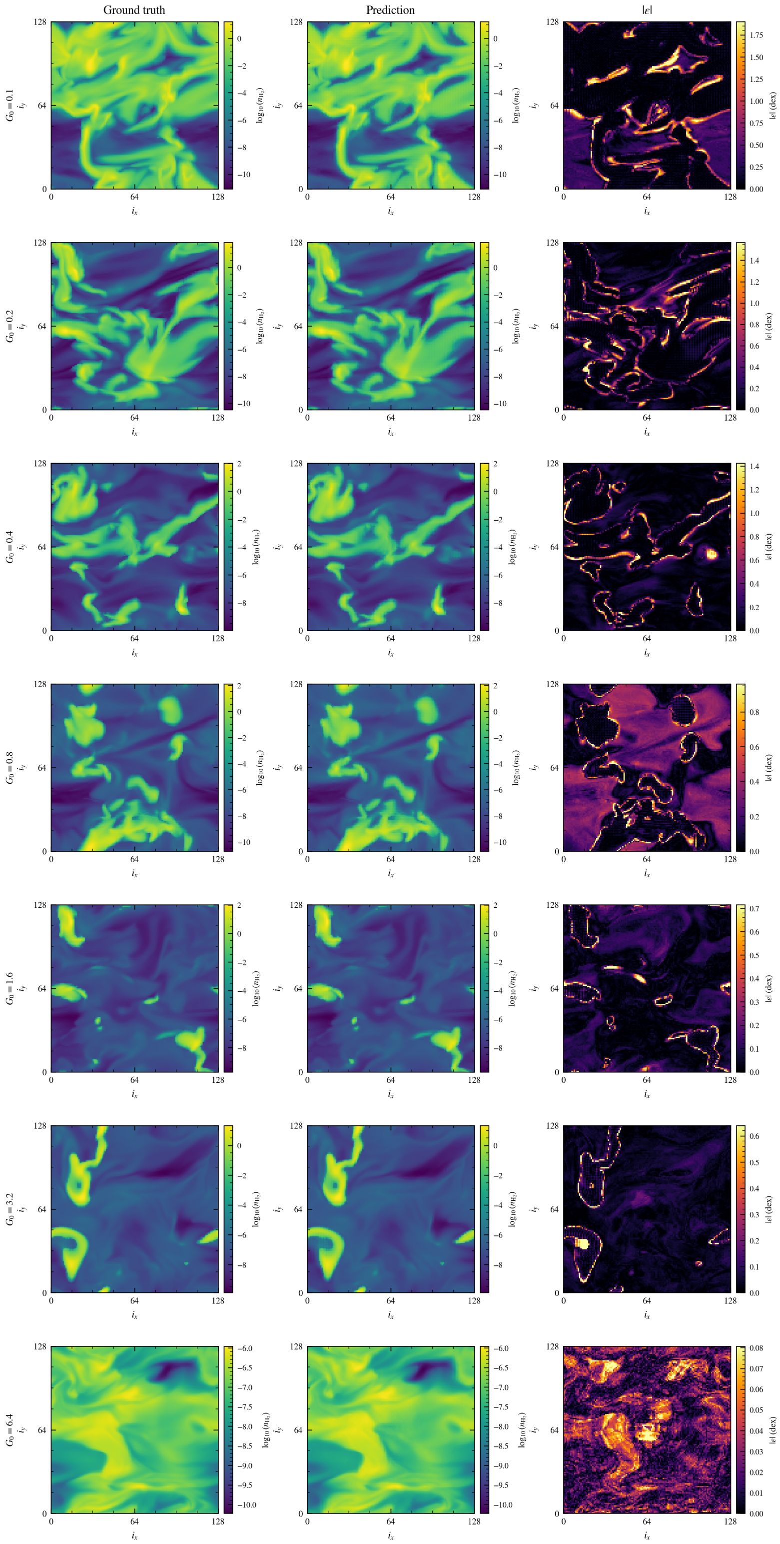

Figure 10. Mid-plane ($z = 64$) slices of the $\log_{10}(n_{H_2})$ field for each G_0 (rows): ground truth, prediction, and absolute error (columns), for the deployed pipeline. Truth and prediction share one colour scale per row; the error column has its own scale. Residuals trace the boundaries of shielded filaments at low G_0; at high G_0 the field is nearly uniform.

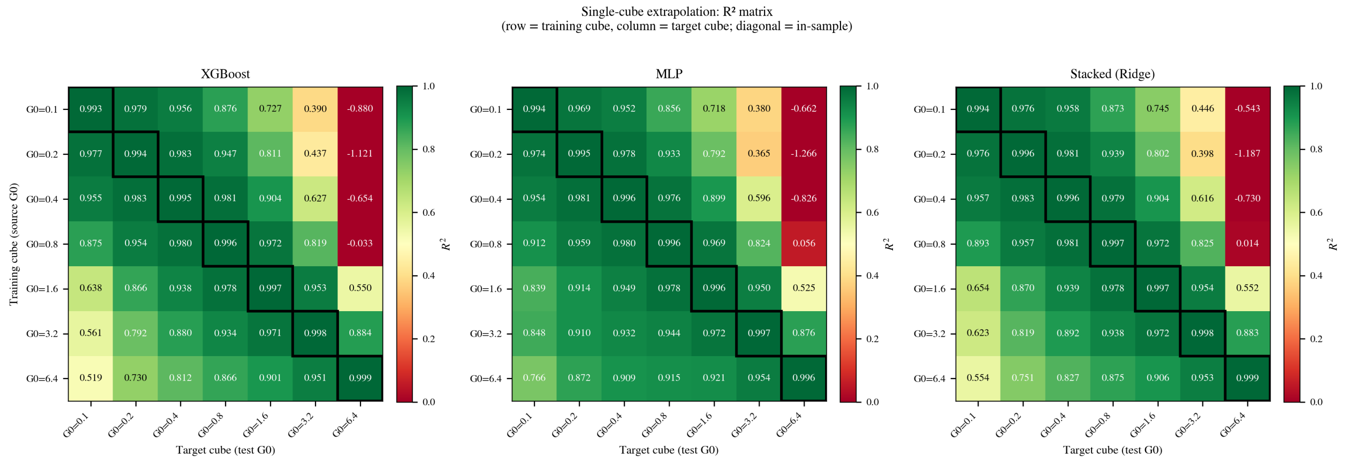

Figure 11. Single-cube transfer: log-space R^2 for models trained on one cube (rows) and evaluated on each cube (columns), for XGBoost (*left*), MLP (*centre*), and the stacked ensemble (*right*). Diagonal cells are in-sample fits. Transfer quality decays with G_0 separation; values for separations of four or more geometric steps are poor and can be strongly negative.

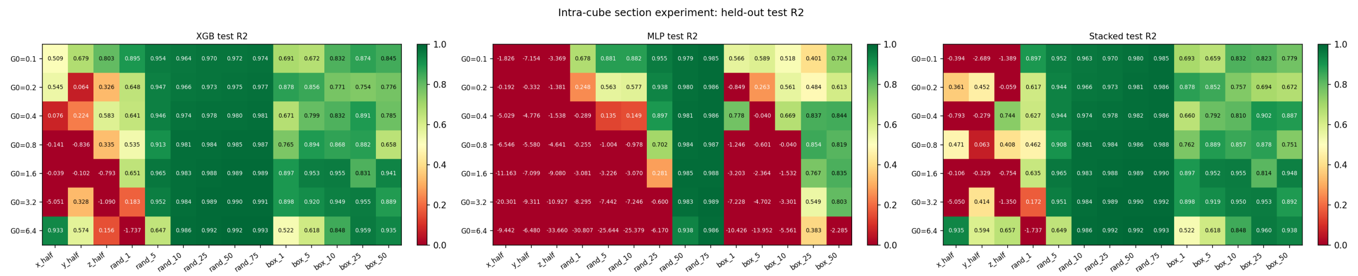

Figure 12. Within-cube sampling geometry: test R^2 per training geometry (columns) and cube (rows) for XGBoost (*left*), MLP (*centre*), and the stacked ensemble (*right*). Random coverage ≥5–10 per cent reconstructs the held-out cells well in every cube; 50 per cent contiguous slabs fail in every cube; compact boxes are intermediate; 1 per cent random coverage is unstable, varying from $R^2 \approx 0.9$ to strongly negative between cubes.

with the broader observation that tree-based models with appropriate features remain highly competitive on tabular problems (Grinsztajn et al. 2022). The bias–scatter decomposition assigns each ingredient a distinct role: spatial features reduce scatter only, calibration relocates bias only (all three calibrated variants of a stack share the identical scatter by construction), and density weighting selects *which* functional of the field is unbiased — three orthogonal levers that compose into the final model. The box-filter means are a deliberately minimal spatial representation; anisotropic kernels aligned with the illumination axis, or cumulative sums along it (a discrete column integral), are natural refinements that we have not exhausted. A physics-based reference point — an analytic molecular-fraction prescription (e.g. Krumholz et al. 2009; Gnedin & Kravtsov 2011; Bialy & Sternberg 2016) evaluated cell-by-cell on the suite — would usefully anchor the ML numbers; published formulae predict equilibrium molecular fractions from column-averaged quantities under geometric assumptions the cell-level task does not satisfy, so a fair comparison requires a calibration step that we leave to future work.

7.2 Volumetric versus per-cell representations

The comparison sharpens into a clean division of regimes. Inside the training range the U-Net is the best model evaluated — interior-fold $R^2 = 0.995$ against 0.984 for the calibrated stack — so the learned volumetric representation demonstrably extracts non-local structure the box filters miss. Outside it, the U-Net fails in a way no global correction can repair: phase-localised, non-monotonic in G_0, with mass-budget violations reaching two orders of magnitude, while the stack's out-of-distribution error is a smooth, two-parameter-correctable offset. The practical reading for limited-data suites is not "tabular beats CNN" but a division of labour: engineered neighbourhood features plus calibration give robust extrapolation; the volumetric model gives the best interpolation and should be preferred only within a densely sampled parameter range. We also note the evaluation-scale dependence of these statements: scoring the same stacked predictions against a $2 \times 2 \times 2$ box-averaged target raises its R^2 from 0.962 to 0.991, i.e. roughly half of the residual error variance lives at the smallest resolved scales, where cell-level structure is partly unpredictable from the available inputs. Applications consuming smoothed fields inherit the higher figure.

7.3 Which model for which use

The metric hierarchy supports use-dependent recommendations rather than a single winner. For physical fidelity of the predicted field — mass budgets, phase statistics — the mass-calibrated weighted stack is the clear choice (RMSE 0.29 dex, mass within 7 per cent, molecular R^2 0.991). For applications weighting all cells equally, the mean-calibrated stack minimises cell-mean error (bias −0.01 dex, skill +0.60) at the price of a 20–50 per cent mass deficit. For generating statistically realistic n_{H_2} fields (synthetic observations, sub-grid realisations) the weighted spatial MLP is attractive: its marginal distribution is the closest to truth in the comparison ($W_1 = 0.109$ dex) because Ridge stacking, which averages predictions, slightly narrows the predicted distribution. And for interpolation-only use within a well-sampled parameter range, the volumetric U-Net is the most accurate option available.

7.4 Deployability and the shielding factor

The ablation of Section 5.3 addresses the most delicate methodological question in this study: whether the surrogate's accuracy is borrowed from the solver through f_{H_2}. The answer is use-dependent. For aggregate statistics the dependence is modest: the solver-independent final model retains $R^2 = 0.964$ with a closed mass budget, because neighbourhood averages of the deployable column-type fields substitute for much of the shielding information. For per-cell accuracy inside the molecular phase the dependence is strong (molecular R^2 $0.991 \rightarrow 0.840$, and 0.70 in the most molecular cube): resolving *which* cells cross the atomic-to-molecular transition requires genuine shielding information. Deployment therefore has a spectrum of options: with a cheap approximate shielding estimate (e.g. from extinction), accuracy should fall between the two blocks of Table 4; with no shielding estimate at all, the without-f_{H_2} block applies directly. What our data cannot quantify is the effect of using an *inconsistent* shielding estimator (one with different systematics from the solver's), which would require forward-modelling the estimator itself.

7.5 Limitations

Dataset scope. The seven simulations grow from one turbulent initial realisation and differ only in their imposed G_0, so their structures are strongly correlated even though the evolved fields are not identical (Section 2.1). All generalisation claims are therefore claims about the UV-field axis; transfer across turbulence realisations, metallicities, dust models, or resolutions is untested and plausibly harder. The suite is also quasi-equilibrium and uniformly illuminated; time-dependent or anisotropic radiation fields would invalidate the z-preserving augmentation and may change the role of the spatial features.

Simulation provenance. This paper characterises surrogate methodology on a controlled private suite, and its conclusions are internal to that suite (relative model performance, feature ablations, sampling geometry). Application of the trained surrogate to scientific inference requires the full simulation specification — code, chemical network, box size, resolution, metallicity, dust-to-gas ratio, driving — which is not part of this study.

Boundary treatment. The spatial features reflect neighbourhoods at the box faces and the U-Net zero-pads its convolutions; cells within three voxels of a face (≈13 per cent of the volume at the largest kernel) therefore receive approximate context. Matching the treatment to the simulations' true boundary conditions is a straightforward refinement we have not evaluated.

CNN comparison. The U-Net was trained with an unweighted loss and an unbounded output, receives per-cube input normalisation at test time through its instance-normalisation layers, and its reported checkpoint is selected on held-out-volume loss (worth ≤12 per cent in RMSE; Section 4.3.3) — a mixture of disadvantages and advantages relative to the tabular protocol. A density-weighted, range-bounded U-Net would be the fairer mass-budget comparison and is left to future work; the configuration-sensitivity results suggest any single U-Net number carries a wide error bar regardless.

Calibration hyperparameters. The density-weighting anchors (q_{99}, $q_{99.99}$, $\alpha = 100$) are a designed choice whose sensitivity we have not mapped; the calibration itself is a two-parameter linear model in $\log G_0$ and inherits the smoothness assumption it encodes. The residual +0.16 dex bias of the deployed model at $G_0 = 0.8$ (Section 5.5) marks the limit of what a global linear correction captures on this suite.

8 CONCLUSIONS

We evaluated machine-learning surrogates for the molecular hydrogen number density in a controlled suite of seven 128^3 UV-only ISM chemistry simulations under leave-one-G_0-out cross-validation, scoring every model on the same native 128^3 cells with a metric hierarchy built around the physical uses of the predicted field. The main conclusions are:

(i) How a surrogate is scored determines what one learns about it. Models separated by $\lesssim 0.03$ in log-space R^2 differ by factors of several in recovered H_2 mass; the minimum-squared-error stacked ensemble carries a +0.32 dex held-out bias (mass over-prediction up to 80 per cent) that R^2 does not register. RMSE decomposed into bias and scatter, the mass budget, and phase-conditional errors should be standard reporting for field-level emulators.

(ii) The ingredients of the final model act on distinct error components: multi-scale box-filter features (computed in ≈30 s) cut *scatter* by roughly a third; density-weighted training closes the pointwise models' *mass budgets* at zero R^2 cost; and a two-parameter, mass-weighted bias calibration in $\log G_0$, fitted on training cubes alone, removes the stack's *bias*. The deployed pipeline reaches RMSE = 0.25 dex, $R^2 = 0.990 \pm 0.006$, total H_2 mass within 9 per cent, and molecular-phase $R^2 \geq 0.98$ on every held-out fold.

(iii) The choice of calibration functional is a physical decision, not a technicality: centring the cell-mean error overshoots the mass budget by 20–50 per cent because the surrogate's residuals are density-dependent; centring the mass-weighted residual closes the budget at the cost of a +0.09 dex diffuse-phase offset. We recommend reporting both.

(iv) The solver-derived self-shielding factor is dispensable for morphology but not for molecular-phase chemistry: without it the final model retains $R^2 = 0.964$ and a closed mass budget, but its molecular-phase R^2 falls from 0.991 to 0.840 (0.70 in the most molecular cube) and its RMSE grows by 60 per cent. A U-Net given *no* chemistry-adjacent inputs at all still reaches $R^2 = 0.893$: the H_2 morphology is largely inferable from dynamics, thermodynamics, and the UV strength, while per-cell chemical accuracy is not.

(v) The volumetric 3D U-Net is the best interpolator in the comparison (interior-fold $R^2 = 0.995$, beating every tabular model) and the worst extrapolator: at the boundary folds it falls below the pointwise XGBoost baseline, with phase-localised biases up to +1.1 dex and mass-budget violations exceeding two orders of magnitude that a global calibration cannot repair — at roughly two orders of magnitude higher training cost. On suites of this size, engineered spatial context plus calibration is the robust default; the volumetric model is preferable only within a densely sampled parameter range.

(vi) The learned local mapping transfers across a factor of ~2–4 in G_0 (single-cube transfer $R^2 = 0.966$ at one geometric step, 0.888 at two) and degrades beyond it, validating leave-one-G_0-out as a non-trivial protocol and confirming that none of the seven simulations is redundant.

(vii) Within a cube, reconstruction quality is controlled by sampling geometry rather than sample volume: 10 per cent uniformly random coverage achieves $R^2 = 0.977$ while 50 per cent contiguous slabs fail ($R^2 < 0$), and at very sparse coverage ($\sim$1 per cent) compact contiguous regions outperform random sampling because the masked spatial features remain exact inside them. These are simulation-space results; translating them to observational survey design would require forward modelling of observables.

The full tabular pipeline — feature construction, training, calibration, and prediction for all seven folds — runs in about an hour on a single consumer GPU, and the trained surrogate predicts all 2,097,152 cells of a new cube in seconds.

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SOFTWARE

Python 3.14.5; PyTorch 2.12 (Paszke et al. 2019); XGBoost 3.2 (Chen & Guestrin 2016); NumPy 2.4 (Harris et al. 2020); SciPy 1.17 (Virtanen et al. 2020); scikit-learn 1.8 (Pedregosa et al. 2011); Matplotlib 3.10 (Hunter 2007). All experiments were run on a single NVIDIA GeForce RTX 3090. Random seeds are fixed per fold; every experiment writes a timestamped JSON log containing its full configuration, per-fold metrics, software environment, and a SHA-256 fingerprint of the input data.

DATA AVAILABILITY

All experiment logs, trained-model outputs, and prediction volumes are archived as timestamped JSON and `.npz` files and are available from the author on request. The simulation suite is not yet public; requests for data access may be directed to the corresponding author.

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